Bending Moment And Shear Force Diagram:- ①

This chapter deals with the shear force and bending moment diagram of statically-determinant beams.

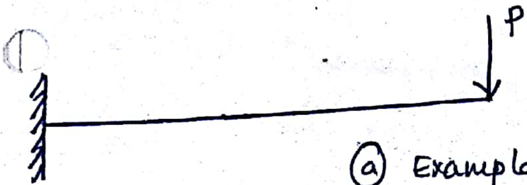
When can we say that the beam is statically determinant,
When the unknown reactions are solved using only the equations of static equilibrium, we can say that the beam is statically determinant.

The equations of static equilibrium are $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$.
When the above equations are satisfied we can say that the beam is under static equilibrium.

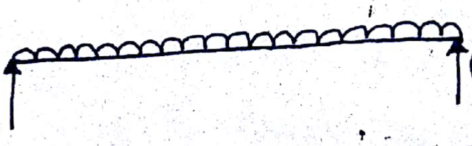
A few examples of statically determinant beams are:-

a) Cantilever beams b) Simply supported beams

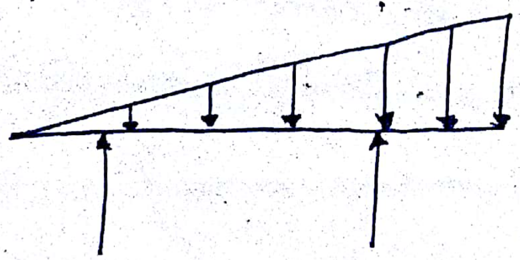
c) Overhanging beams.



ⓐ Example of a Cantilever beam subjected to a point load at the tip.
It could also be that the Cantilever be subjected to Uniformly distributed load or Uniformly Varying loads.



ⓑ Simply supported beam subjected to a Uniformly distributed load (UDL)



ⓑ Overhanging beam subjected to Uniformly Varying load.

Let us consider a beam subjected to a force which is equal in magnitude and opposite in direction as shown in the figure.

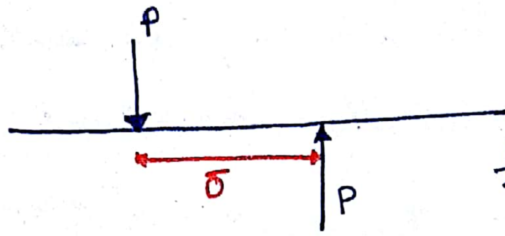
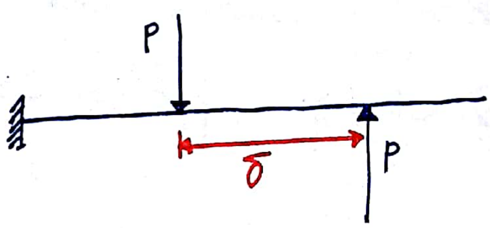
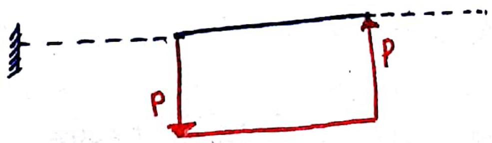


fig. '1'

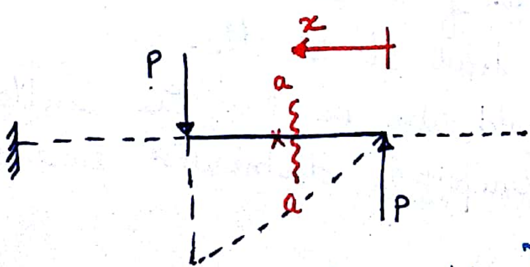
The two forces are at a distance 'δ' apart. These two forces can cause a shearing effect as they are equal and opposite with a small distance between them.

As, we can see the above fig '1.' is not in equilibrium because of the moment the two forces create due to 'δ'. To bring this system to equilibrium we do the following.





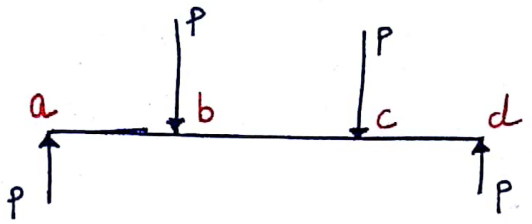
Drawing shear force diagram between the forces.



Drawing bending moment between the forces.

As we can see the moment at section 'a-a' is $P \cdot x$ and the moment in the entire section of length 'δ' is $P \cdot \delta$, but if δ is very small then we can approximate $P \cdot \delta \simeq 0$. Hence, we can say the above system is in 'pure shear'.

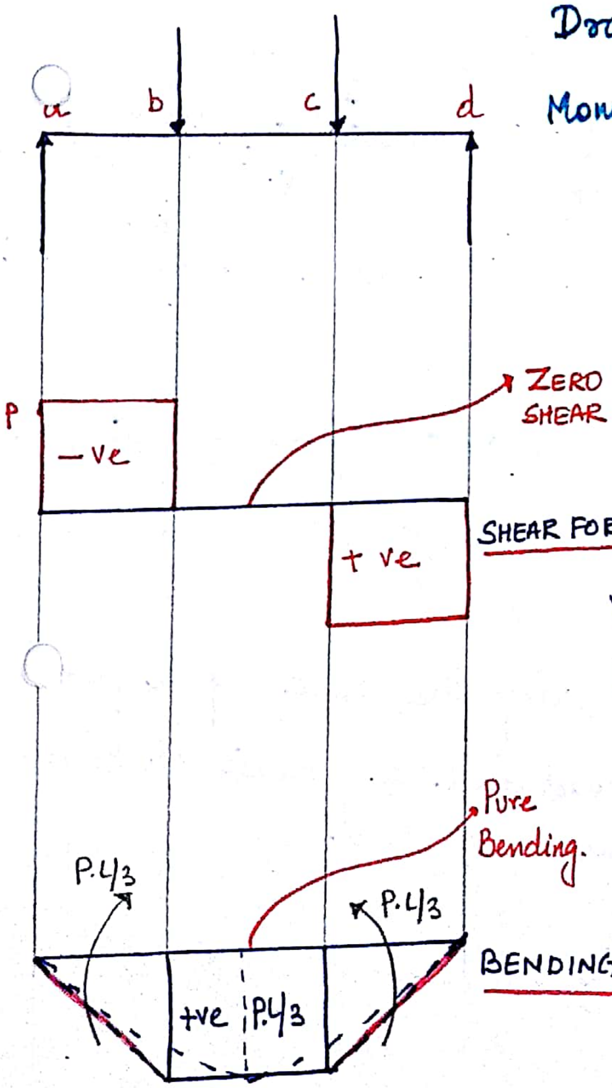
Let us Consider the following System as shown in the figure.



The reactions at the simply supported ends will be 'P' due to two forces P acting transversely to the axis of the beam.

The distance between the reaction and forces be L/3 where 'L' is the total length of the beam.

Drawing the Shear force & bending Moment diagram.



To obtain the shear force diagram at 'a' we get P going upwards and continous till 'b' where at 'P' force at b drops the line back to the beam. and similarly ; we can get the SF-diagram.

SHEAR FORCE DIAGRAM.

To obtain the bending moment diagram we can replace the forces and reaction between 'a & b' by a Couple which will be P.L/3 and similarly at c and d. Hence, we have a pure shear situations and pure bending situation's over the beam.

BENDING MOMENT DIAGRAM.

Sign Convention: We follow the standard Sign Convention as mentioned below.

+ bending moment (Sagging)

-ve bending (Hogging)

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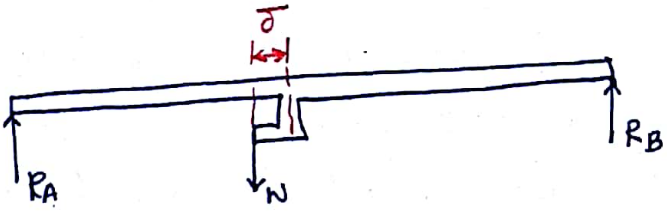
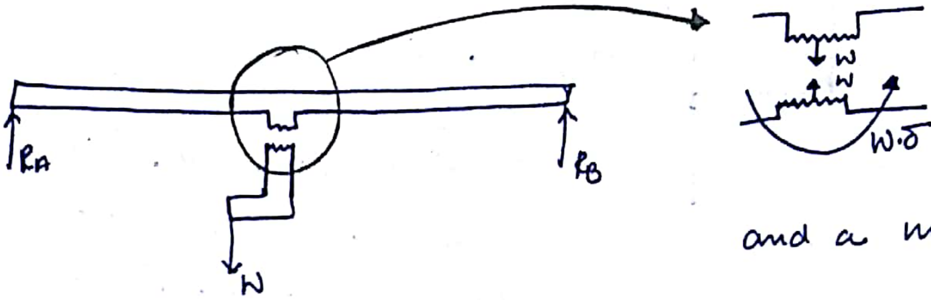
Special loading Cases in beams :-

(1)

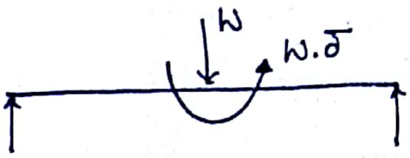


Now for the above Case we can replace the hook at the Centre of the beam with a point load or Concentrated load and a moment.



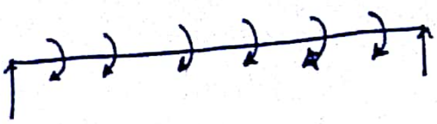
Reaction forces W due to hook.
and a moment W·δ.

Hence, we can draw an 1·D diagram replacing the hook on the beam.



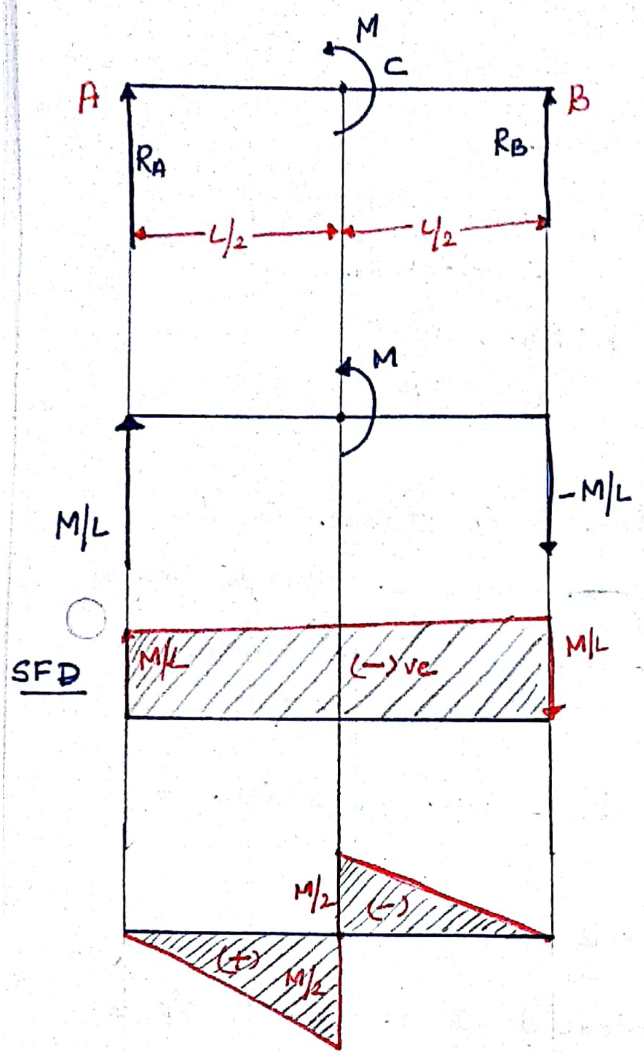
Replacing the hook by a point load and a counter-clockwise moment W·δ.

(2) The other special loading Case is,



Uniformly distributed moments.

Example: 1. A beam subjected to a anti-clockwise moment at the centre of the beam. 5



To calculate reactions R_A & R_B,

We take moment at R_B.

$\Sigma M_B = 0$,

$R_A \times L - M = 0$, $R_A \cdot L = M$

$R_A = M/L$ (+ve)

Similiarly,

ΣR_A(moments) = 0 , $\Sigma M_A = 0$

$-(R_B \times L) - (M) = 0$

$R_B \times L + M = 0$

$R_B = -M/L$

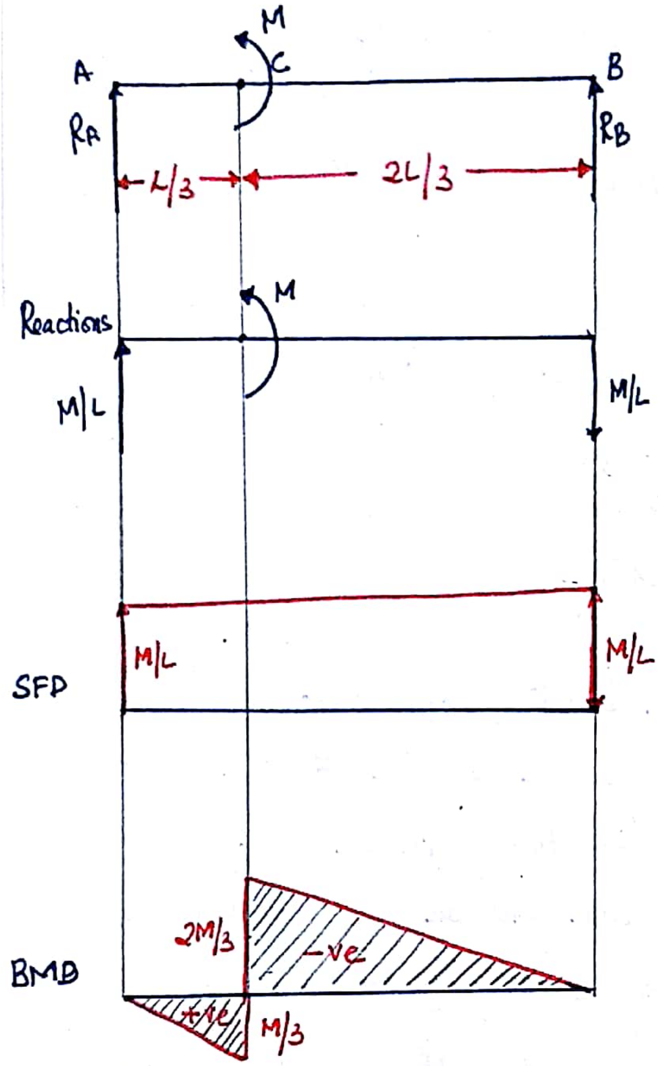
'at

At 'C', the bending moment due to M at the Centre will be $M/L \times L/2$ and its direction will be +ve (clockwise). The maximum will be at 'C', and linearly decreases as it gets closer to 'A'.

Similiarly, at the other section 'C-B'. We can see that the moment due to $-M/L$ at C is $-M/L \times L/2$. Hence, the moment at C due to R_B is $-M/2$, which is again maximum at C and linearly decreases to 'B'.

The sign convention at A-C is positive as $M/2$ is positive and the sign at C-B is -ve hence, the change in sign at 'C'. Note that the value of bending moment is 'M' at C which indicates that the bending moment is maximum at 'C'.

Example: 2 :-



$\Sigma M_B = 0$, $R_A \times L - M = 0$

$\therefore M/L = R_A$

$\Sigma M_A = 0$, $-R_B \times L - M = 0$

$R_B = -M/L$

Note, There is no change in the shear force diagram. due to change in the location of the moment.

Moment at C due R_a, which is M/L is

$M_c = \frac{M}{L} \times \frac{L}{3} = M/3$

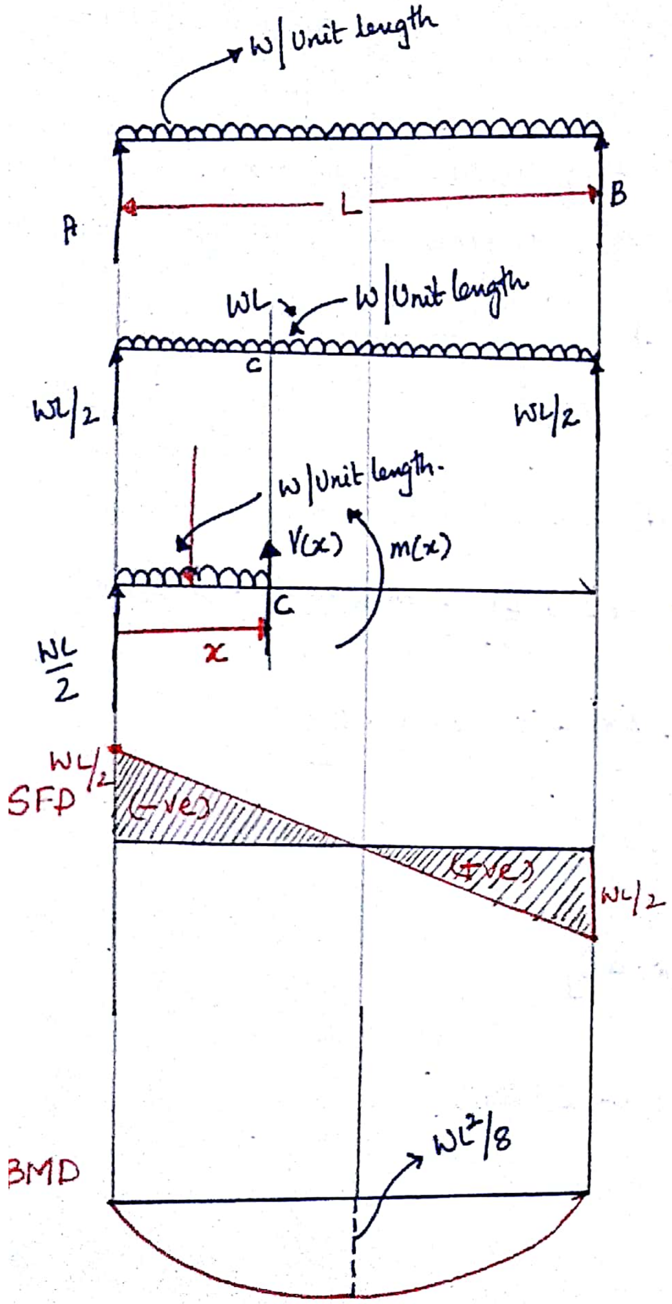
and Moment at C due to R_B is

$M_c = \frac{M}{L} \times \frac{2L}{3} = M2/3 = 2M/3$

Note that in Section 'A-C' the bending moment is positive and in the section 'C-B' the bending moment is -ve.

Keep in mind that when the value of M at C is added we should get back the value of the applied moment.

Example 3:- Simply Supported beam Subjected to UDL or Self Weight.



Let us assume that Self weight of the beam is w/unit length.
Hence, the complete weight acting will be W.L where L is the length of the beam.
Hence due to symmetry we can say that the reactions are WL/2 at A and B.

Let us take a section at distance 'x' from A.
And the shear force at the section will be V(x) and reaction at A is WL/2. and bending moment at section is m(x).
Taking the equation of Static equation

$\Sigma F_y = 0$, we get,

$$\frac{WL}{2} + V(x) - W.x = 0$$

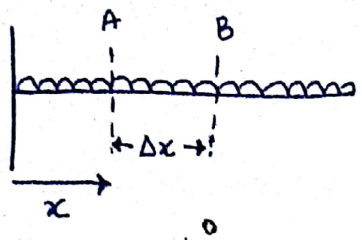
Hence, $V(x) = W.x - \frac{WL}{2} = W(x - L/2)$ — (1)

∴ We can draw the shear force as a function of (x). We can note that at $x = L/2$ the shear force is zero. and max at $x=0$, where shear force is '-ve'.
To draw BMD, we do, $\Sigma M_C = 0$, ⇒ $\frac{WL}{2}.x - m(x) - \frac{Wx}{\cancel{2}}.\frac{x}{2} = 0$

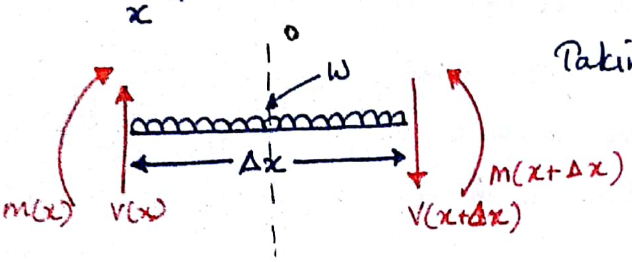
$m(x) = -\frac{Wx}{}.\frac{x}{2} + \frac{WL.x}{2}$, $\frac{W}{2}(L-x)x = m(x)$ — (2)

Hence, we can see that at $x=0$, $m(x) = 0$ and at $x = L$ $m(x) = 0$,
So, we can say that bending moment is zero at the ends. And the equation is quadratic. Hence, at mid span where $x = L/2$, we have
$m(x) = WL^2/8$ //

Relationship between shear force and bending Moment.



Let us Consider a Section 'A-B' at a distance 'x' from the left end with length (small) Δx.



Taking equations of static equilibrium.

$$\Sigma F_y = 0, \quad V(x) - V(x+\Delta x) - w \cdot \Delta x = 0$$

rearranging. $w = \dfrac{V(x) - V(x+\Delta x)}{\Delta x}$

Applying Limits $\Delta x \to 0$, we know, $\boxed{\dfrac{dv}{dx} = w(x)}$ — (1)

Thus variation of shear force is function of $w(x)$.

Similarly taking $\Sigma M_0 = 0$, $V(x) \cdot \dfrac{\Delta x}{2} + V(x+\Delta x) \cdot \dfrac{\Delta x}{2} - M(x+\Delta x) + m(x) = 0$

rearranging,

$$M(x) - m(x+\Delta x) = -V(x)\frac{\Delta x}{2} - V(x+\Delta x) \cdot \frac{\Delta x}{2}$$

$$\therefore \frac{m(x) - m(x+\Delta x)}{\Delta x/2} = -[V(x) + V(x+\Delta x)]$$

or $\dfrac{[m(x) - m(x+\Delta x)]2}{\Delta x} = -[V(x) + V(x+\Delta x)]$

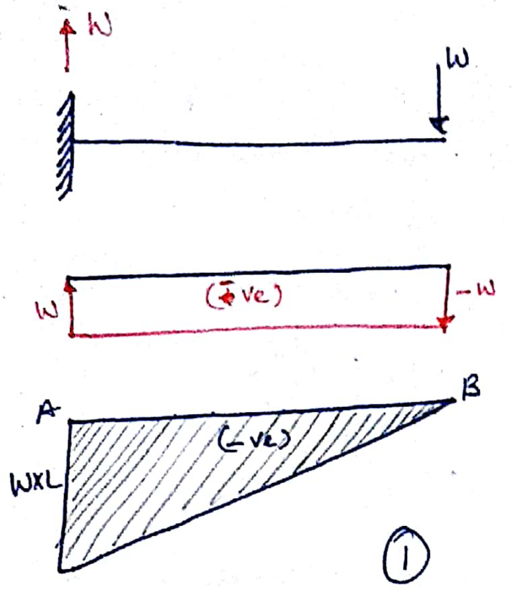
When $\Delta x \to 0$, we have, $2\dfrac{dm}{dx} = -2V(x)$

$$\therefore \boxed{\frac{dm}{dx} = -V(x)} \quad — (2)$$

From (1) and (2) $\dfrac{d(-dM/dx)}{dx} = \boxed{\dfrac{d^2m}{dx^2} = -w(x)}$

Hence, we can conclude that when 'V' is Constant, 'm' is linear and when 'V' is zero, 'm' is Constant. Or otherwise we can say, 'm' is one degree higher than V. Hence in example '3' where shear force is linear, bending moment is quadratic.

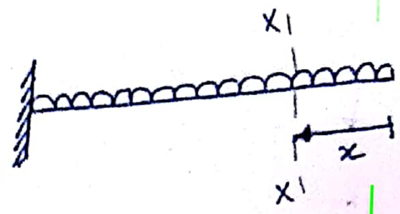
Some more quick examples:



$V = -W$

Constant shear force

Linear B.M.



Taking moment at x-x

$M = Wx \cdot \frac{x}{2} = \frac{Wx^2}{2}$

or when $x = L$.

$M = \frac{WL^2}{2}$

and when $x = 0$, $M = 0$

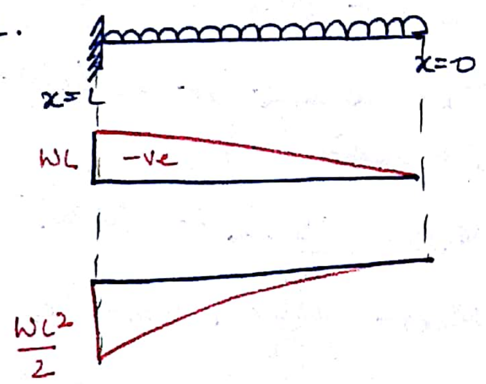
Similarly for shear for

$V(x) = W \cdot x$

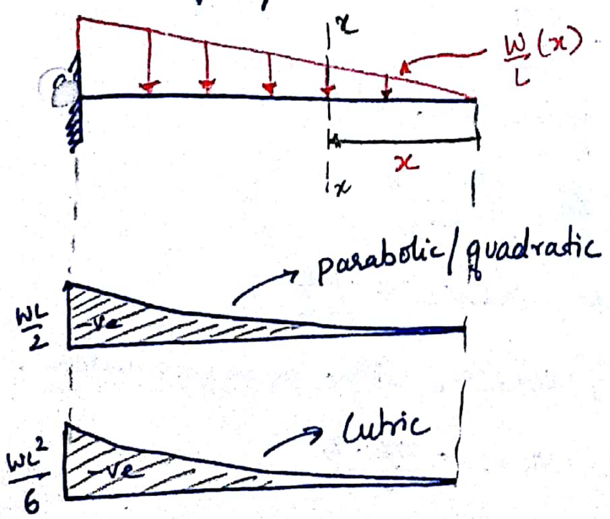
when $x = 0$, $V(x) = 0$

and $x = L$, $V(x) = W \cdot L$

hence.



Similarly for Cantilever with Uniformly distributed load.



At section x-x, we know,

$F = \frac{Wx}{L}$ hence, linear force, hence the shear force will be quadratic and bending moment will be Cubic.

$V = \frac{Wx}{L} \cdot \frac{x}{2} = \frac{Wx^2}{2L}$, $x = 0$, $V = 0$

$x = L$, $V = \frac{WL}{2}$ ~~at~~

but the variation is quadratic.

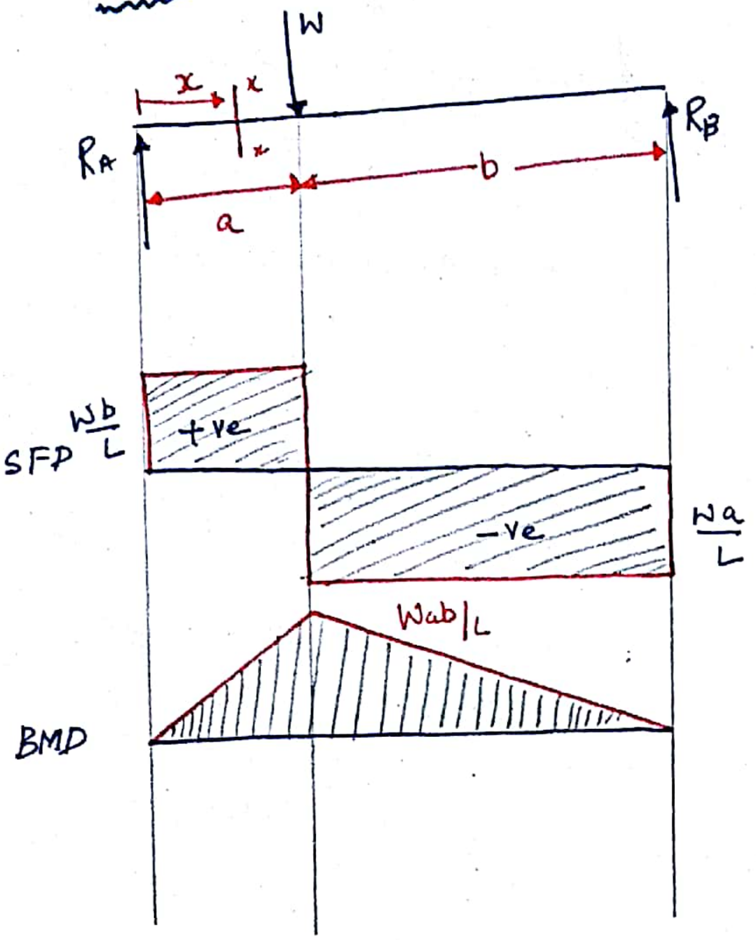
and bending moment is

$M = \frac{Wx}{L} \cdot \frac{x}{2} \cdot \frac{x}{3} = \frac{Wx^3}{6L}$ Cubic

Variation, $x = 0$, $M = 0$

$x = L$, $M = WL^2/6$

Simply Supported beam Subjected to Concentrated load.



To obtain reactions we take moment at (A).

$\Sigma M_A = 0, \quad + W \times a - R_B \times L = 0$

$\therefore R_B = \frac{Wa}{L}$

Similiarly $\frac{Wb}{L} = R_A$

Since, SFD is linear/Constant BMD will be linear with moment being.

Take moment at x between the region 'a'.

$M(x) = \frac{Wb}{L} \cdot x + W(a-x)$

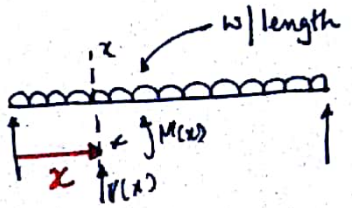
$M(x) = \frac{Wbx}{L} + Wa - Wx$

$M(x) = \frac{W}{L}\left[bx + aL - xL\right]$

Substituting $x = a$, we get, $\frac{W}{L}\left[ba + \cancel{aL}^{0} - \cancel{aL}^{0}\right]$

hence, $M(x) = \frac{Wba}{L}$ at distance 'a' from the LHS.

Simply Supported beam subjected to UDL :-



$V(x) \Rightarrow \frac{WL}{2} - Wx + V(x) = 0, \quad V(x) = Wx - \frac{WL}{2}$

hence, when $x = L/2$, $V(x) = 0$, with linear Variation.

Moment $M(x) \Rightarrow \frac{WL}{2} \cdot x - M(x) - \frac{Wx}{2} \cdot \frac{x}{2}$

$\therefore M(x) = \frac{WL}{2} \cdot x - \frac{W \cdot x \cdot x}{2}$

Solved as example (3) in page (7) :-

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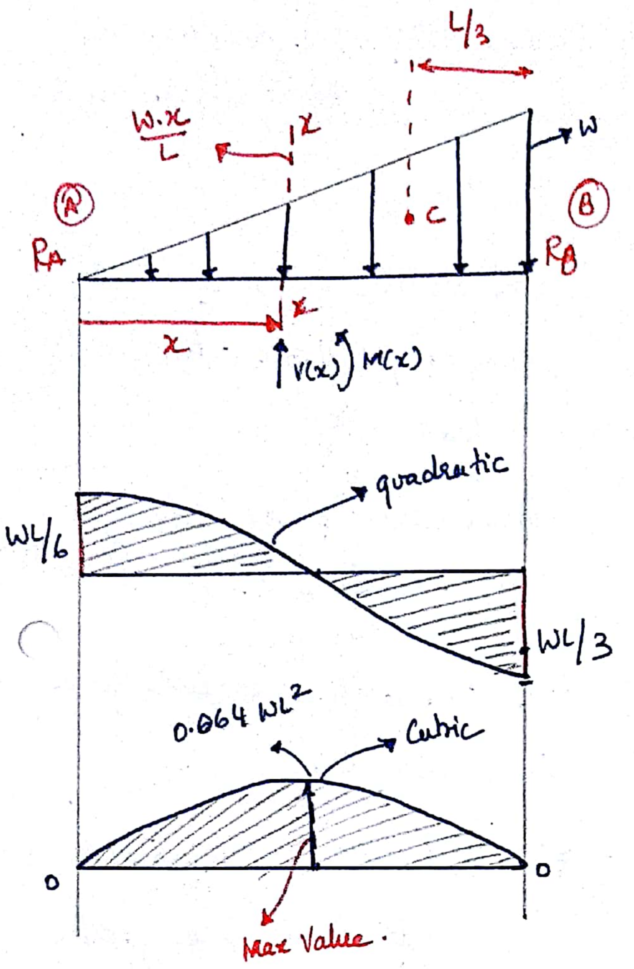
Simply Supported Beam Subjected to UVL.



Calculation of Reactions R_A & R_B

Total load $= \frac{1}{2} \cdot L \cdot W = \frac{WL}{2}$

acting at C.

$\therefore \Sigma M_A = 0, \quad \frac{WL}{2} \cdot \frac{2L}{3} - R_B \times L = 0$

$R_B = \frac{WL^2}{3} \cdot \frac{1}{L} = \frac{WL}{3}$

and similiarly $R_A = \frac{WL}{6}$

To Calculate shear at x-x,

$\Sigma F_y = 0$ at (x-x to 0)

$+\left(\frac{W.x}{L} \times \frac{x}{2}\right) + V(x) - \frac{WL}{6} = 0$

$V(x) = \frac{WL}{6} - \frac{Wx^2}{2L}$

when $x=0$, $V(a) = \frac{WL}{6}$ and

when $x=L$, $V(b) = \frac{WL}{6} - \frac{WL}{2} = -\frac{WL}{3}$.

Hence, when can see a parabolic distribution.

at $x = ?$ the sign changes or $V(x) = 0$, $\therefore V(x) = 0 = \frac{WL}{6} - \frac{Wx^2}{2L}$

$\frac{WL}{6} = \frac{Wx^2}{2L} \qquad x^2 = \frac{1}{3}L^2 \qquad x = 0.577\,L$ the sign changes.

$\therefore$ To calculate moment at x-x, $R_A \cdot x - \frac{Wx^2}{2L} \cdot \frac{x}{3} - M(x) = 0$

$M(x) = \frac{WLx}{6} - \frac{Wx^3}{6L}$ (Cubic Variation)

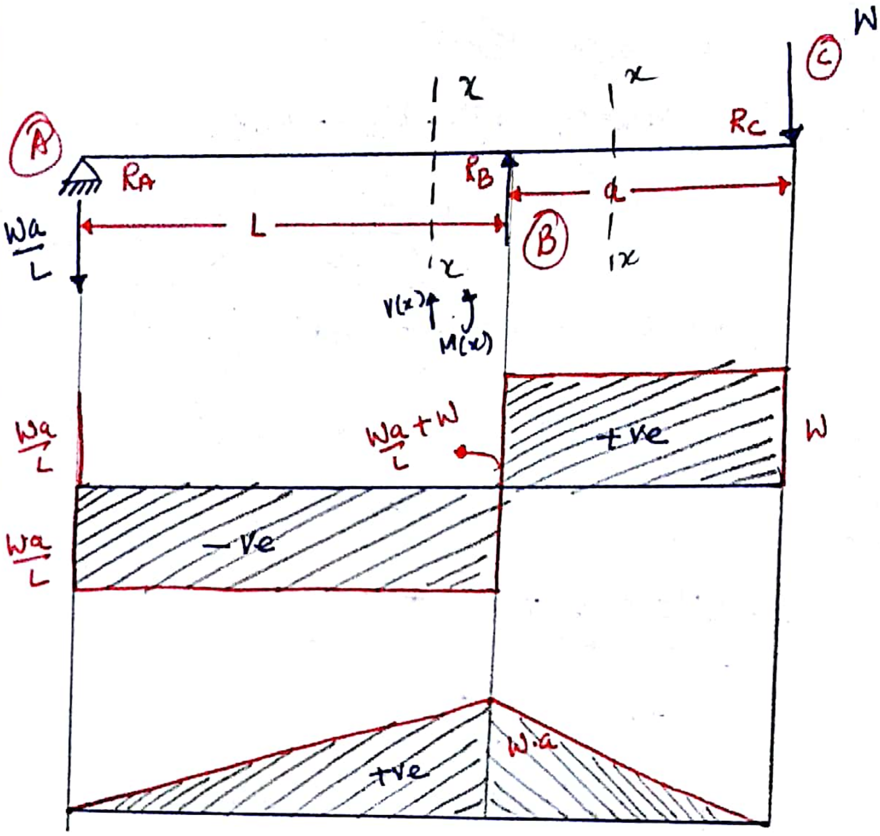
$M(x) = \frac{WLx}{6} - \frac{Wx^3}{6L}$, when $x = 0$, $M(x) = \frac{\cancel{WL}}{6} = 0$ at A

and when $x = L$, $\frac{WLx}{6} - \frac{WL^3}{6L} = \cancel{\frac{WL^2 - WL^2}{6}}\ \frac{WLx}{6} - \frac{WL^3}{6L} = \frac{WL^2}{6} - \frac{WL^2}{6} = 0$ // at B.

differentiating, $\frac{dM}{dx} = 0$, $\left(\frac{WLx}{6} - \frac{Wx^3}{6L}\right) d/dx = 0$, $\therefore$ we get $x = \frac{L}{\sqrt{3}}$ //.

Overhanging beam subjected to Concentrated load at the free end.

Please Note: The reaction at 'B' is downward.



Calculating reactions

$\Sigma M_B = 0, \quad W \times a - R_A \times L = 0$

$R_A = \frac{Wa}{L}$

Similiarly, we can get

$R_B = \frac{Wa}{L} + W.$

we know the shear force is constant,

To calculate moment at x-x we take moment at x-x.

$\Sigma M_x = 0$, taking moment only on the left of 'x-x'

$-\frac{Wa}{L}.x - M(x) = 0$

$M(x) = -\frac{W.a.x}{L}$ (Linear Variation

at $x=0, \quad M(x) = 0$

$x = L, \quad M = W.a$.

Similiarly on section with length. 'a' from the free end.

$W.x = M(x)$

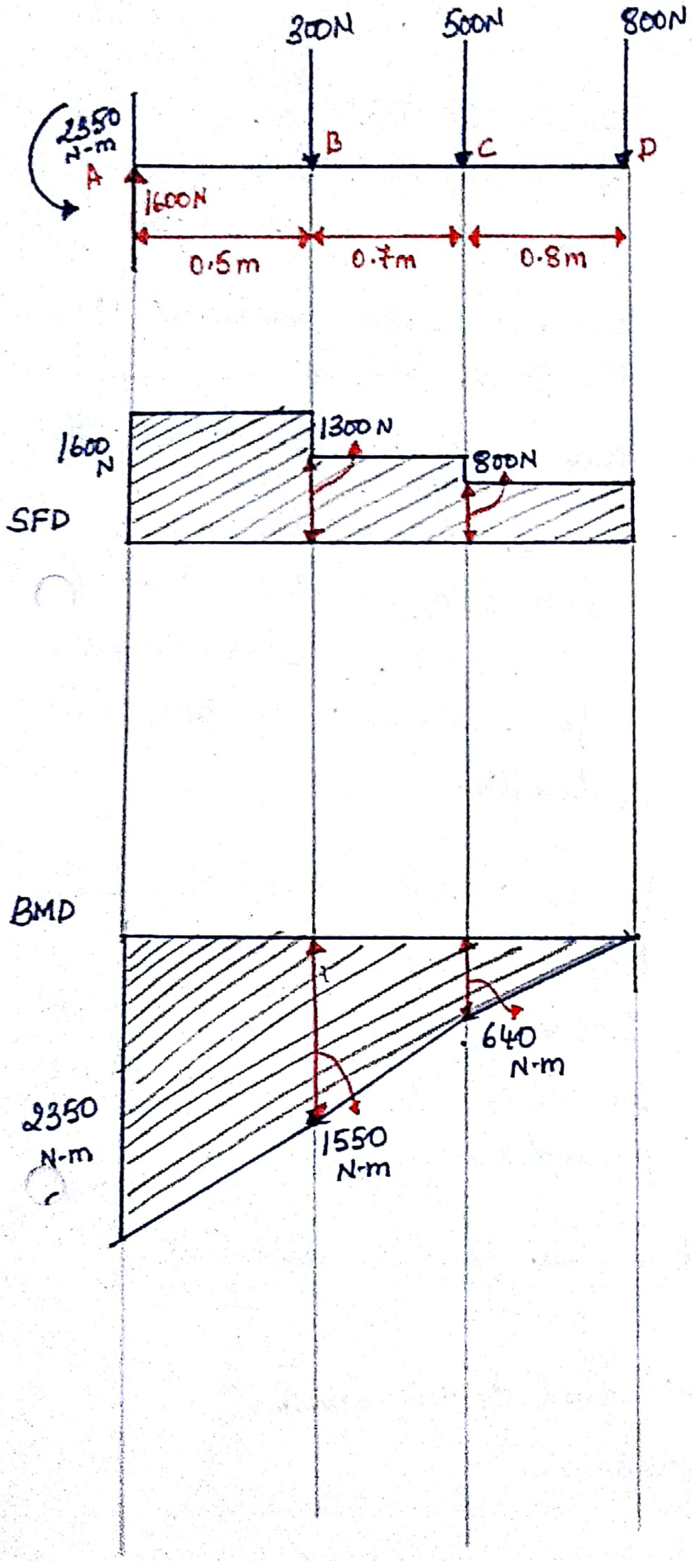
when $x=0, \quad M=0$

$x = a \quad M = Wa.$

Hence, we can plot SFD & BMD

Problem & Examples :-

1.



The load is a point load and hence, the shear force will be Constant and the bending moment will Vary linearly.

The total shear force is

300 + 500 + 800 N = 1600 N

Hence the reaction at (A) will be 1600 N acting upwards

Shear force between AB = 1600 and b/w BC = 1300 and b/w CD = 800 N.

Now, to draw the BMD.

The total bending moment at the fixed end is,

$(-800 \times 2) + (-500 \times 1.2) + (-300 \times 0.5)$

$= -2350$ N-m

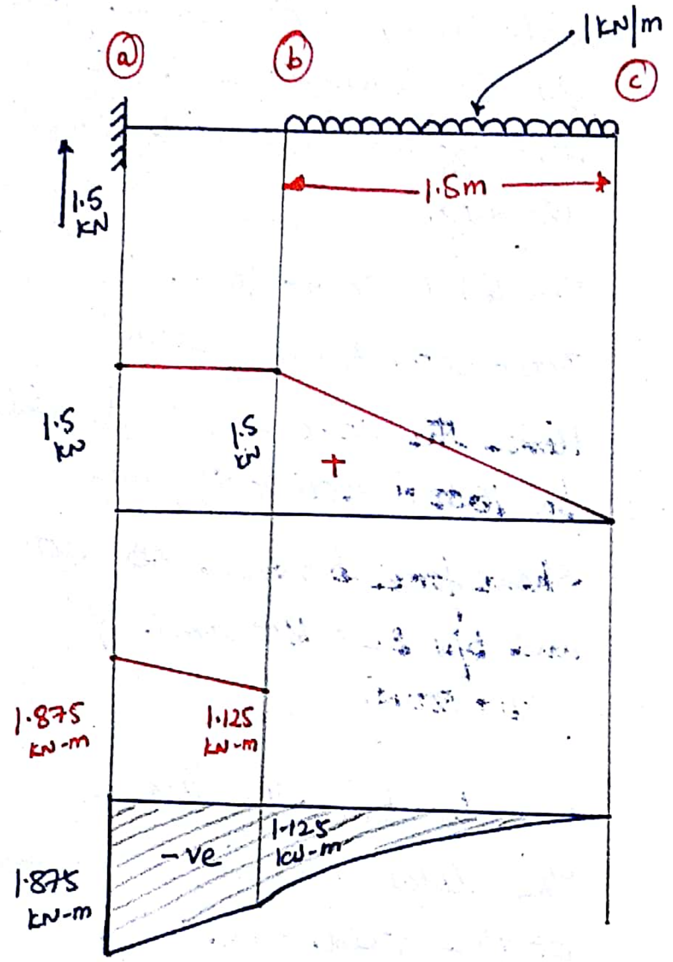
acting anti- clockwise.

Moment between b-c is

$(-500 \times 0.7) + (-800 \times 1.5) = -1550.$

and Moment b/w c-D is

$-800 \times (0.8) = 640$ N-m.



Reaction at the ~~free~~ fixed end is 1.5 kN due to 1 kN/m load over a span of 1.5m.

And the reaction moment at the fixed end is

~~...~~ $-1\ \frac{kN}{m} \times 1.5m \times (0.5 + 0.75) = 1.875$

Where 0.5 is distance of a-b and 0.75 is half of 1.5 because the resultant force will act at half the length.

To draw S.F.D & B.M.D, we know because its a constant load ~~...~~, SFD will be linear and B.M.D will be parabolic.

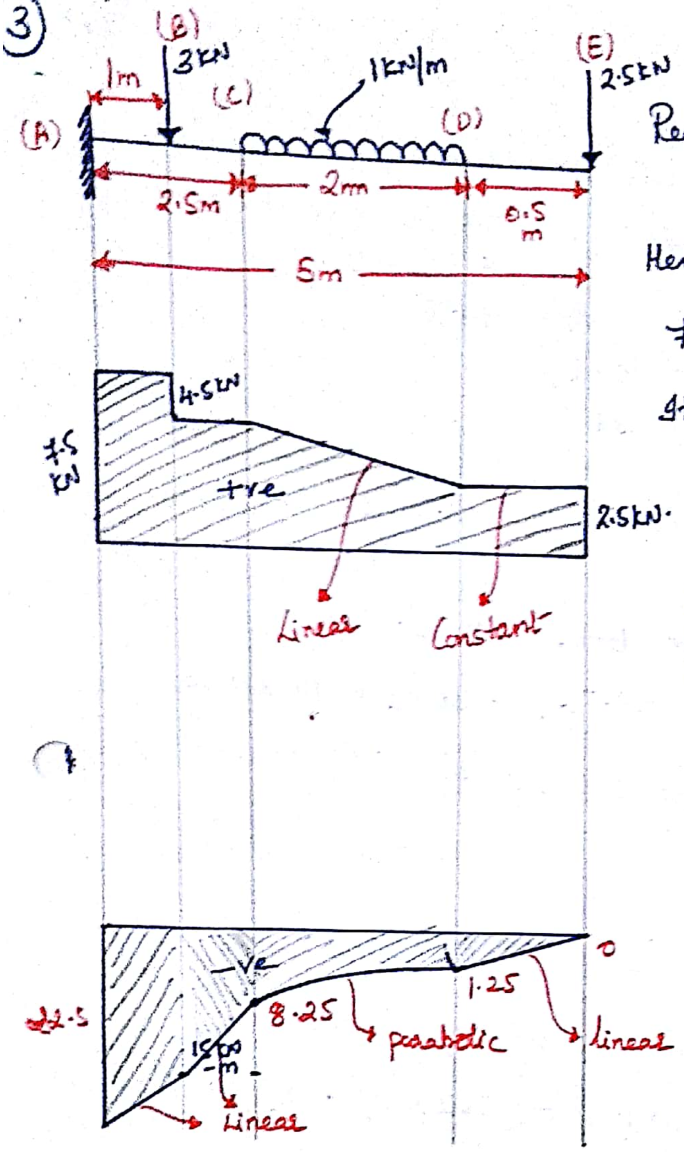
Hence shear force will be zero at free end and so will be B.M.D.

At section 'B' S.F.D will be at 1.5 kN and B.M.D will be

$$W \cdot (1.5) \cdot \frac{1.5}{2} = \frac{1 \cdot (1.5)^2}{2} = 1.125 \text{ kN-m}.$$

Hence, it will be a parabolic variation b/w b & c and linear b/w a & b and similarly for SFD, we know, its linear b/w b & c and constant b/w 'a & b'.

3



Reaction at (A) is $1 \times 2 = 2$
$+ \ 2.5 + 3 = 7.5$ kN

Hence, shear force at the fixed end is 7.5 kN.

It is constant b/w 'a-b', 'B-C' & 'D-E' but linear b/w 'C-D'.

B.M.D.

$(-2.5\times5) + (-2\times3.5) + (-3\times1) = -22.5$ kN-m

at free end. -22.5 kN-m.

Bending Moment a 'b'.

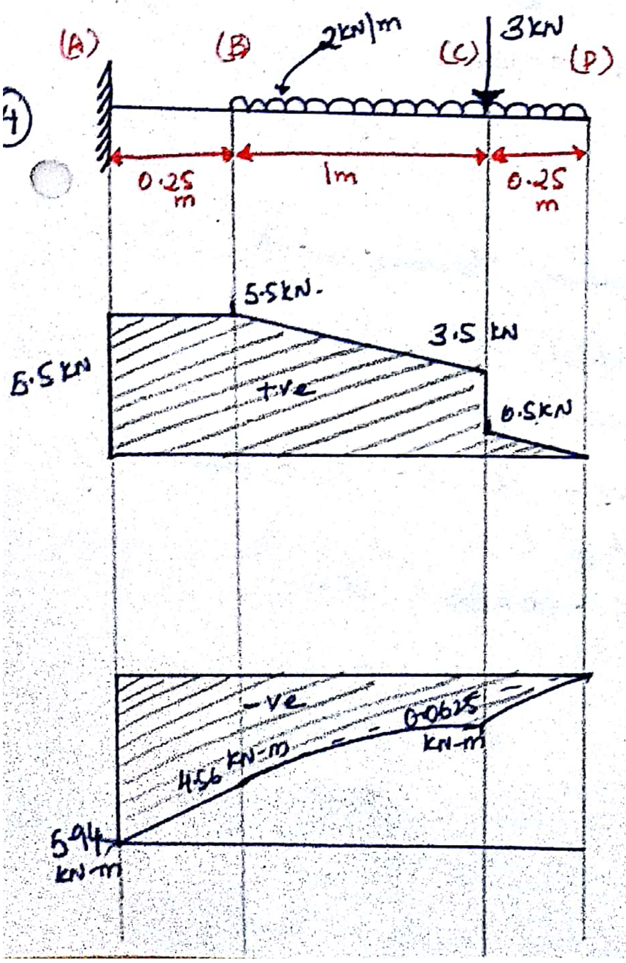
$(-2\times2.5) + (-2.5\times4) = -15$ kN-m (Linear)

B.M at 'c'.

$(-2\times1) + (-2.5\times2.5) = -8.25$ kN-m

B.M @ 'd' $= -2.5\times0.5 = -1.25$ kN-m

4



Reaction at (A) $3 + (2\times1.25) = 5.5$ kN

Linear b/w B-D and Constant b/w 'a-b' &

Bending Moment at (A)

$(-2\times1.25)(0.625+0.25) = -2.19$ kN-m

$+ (-3\times1.25) = -3.75$

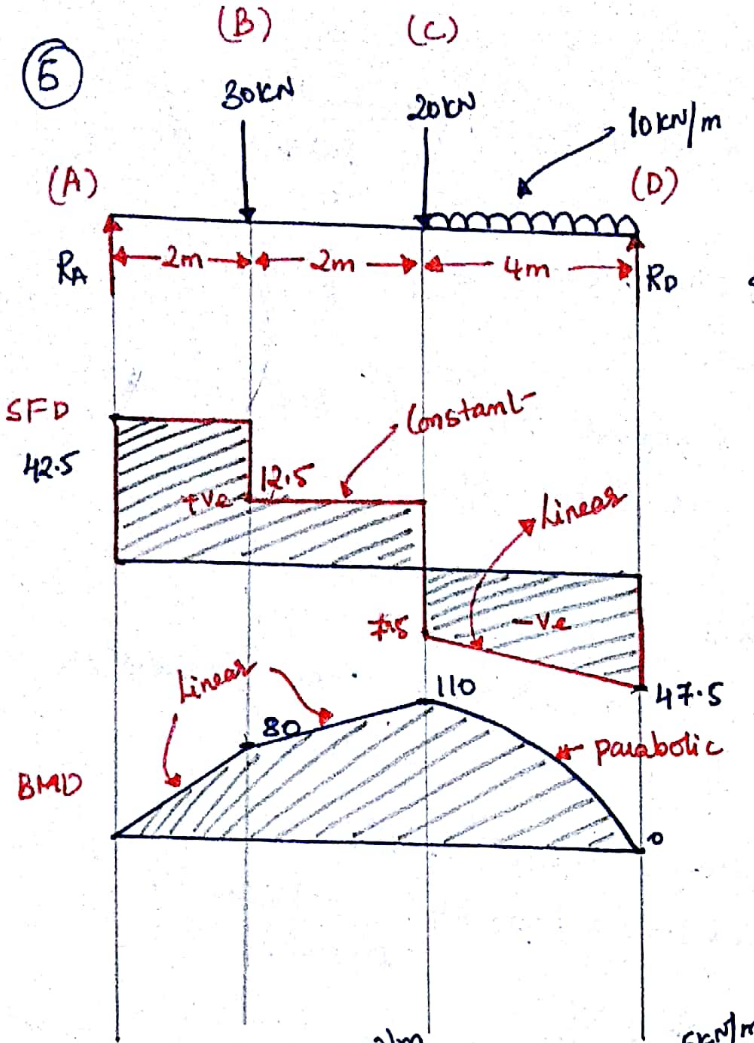
$= 5.94$ kN-m

at B. $(-2\times1.25)(0.625) + 3 = 4.56$ (Linear)

at (c) $(0.25\times2)(0.125)$ parabolic

0.0625

5



Calculating reactions

$\Sigma M_A = 0$, $(30\times2)+(20\times4)+(10\times4\times6) = R_D \times 8$

$R_D = 47.5$

$R_A = 90 - R_D$

$R_A = 42.5$

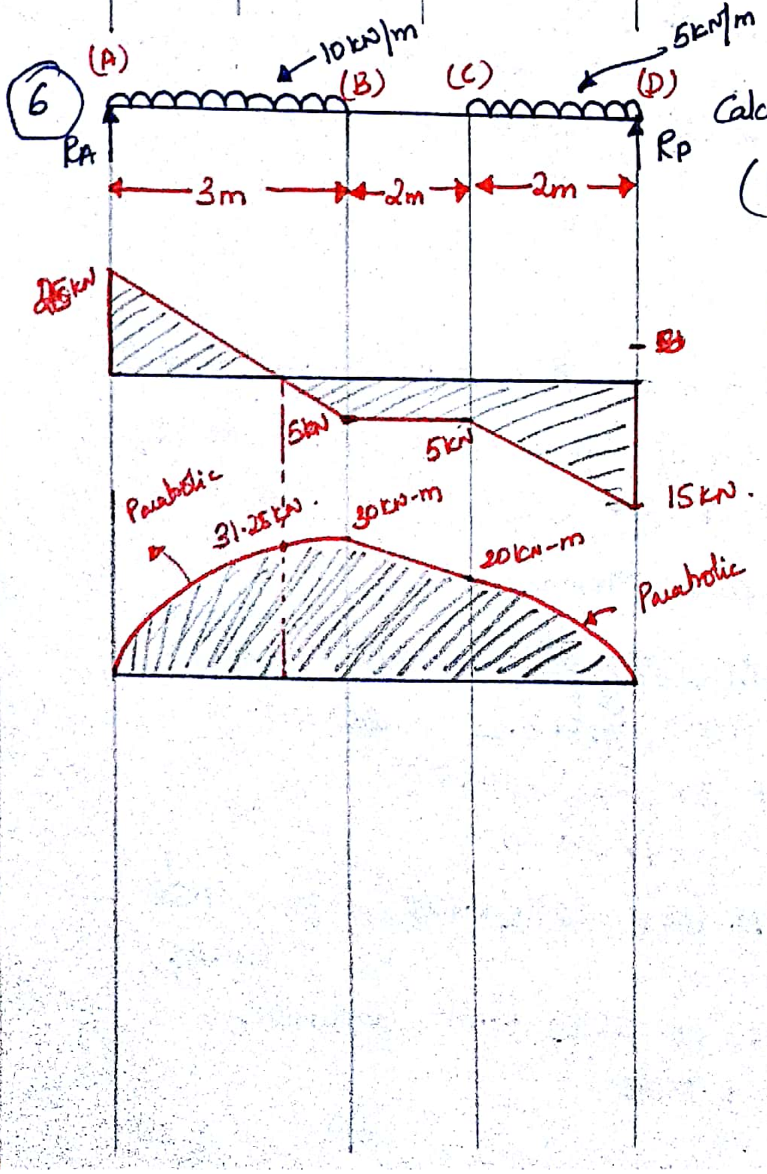
For 'A-B', $42.5\times2 = 80$ kN-m

for 'B-C' at 'C'.

$42.5\times4 - 30\times2 = 110$ kN-m

for C-D at 'D'

6



Calculating reactions, $\Sigma M_A = 0$

$(10\times3\times1.5)+(5\times2\times6) = R_D\times7$

$R_D = 15$ kN

$R_A = 40$ kN $- 15$ kN

$R_A = 25$ kN

Bending Moment at (A) = 0

$\cancel{(10\times3\times1.5)+(5\times2\times6)/7}$

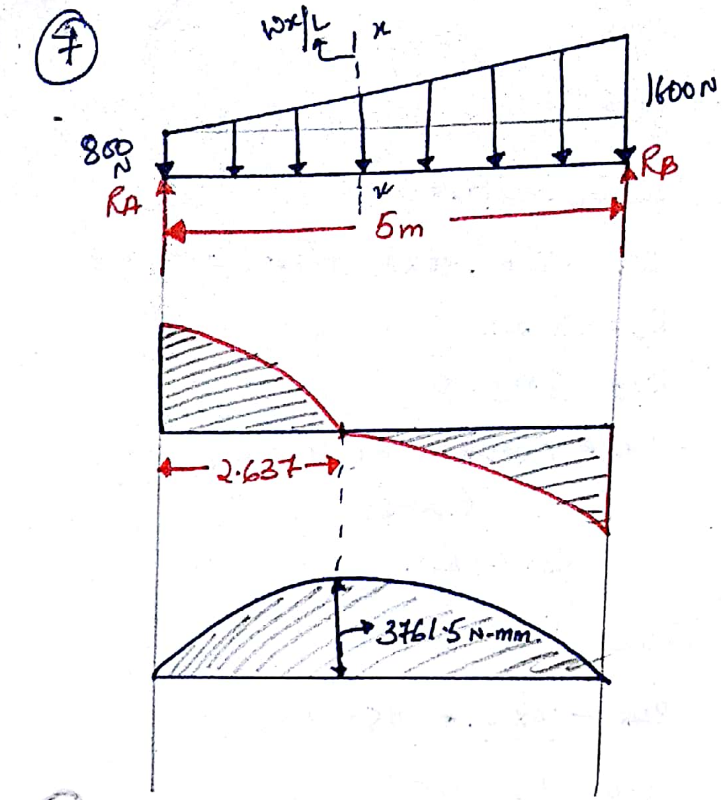
B.M @ B.

$25\times3 - 30\times1.5 = 75 - 30$ kN-m

B.M @ C

$25\times5 - 30\times3.5 = 20$ kN.

7



Reactions at the ends are:-

$800 \times 5 = 4000N.$

$+ \ 800 \times 5 \times \frac{1}{2} = 2000N$

$R_A + R_B = 6000N.$

Moment at A $\Sigma M_A = 0$

$4000 \times 2.5 + 2000 \times \frac{2}{3}(5) = R_B \times 5$

$R_B = 3333.33\ N.$

$R_A = 2666.66.$

Bending Moment @ x-x

$R_A \times x - 800 \times x \cdot \frac{x}{2} - \frac{wx}{L} \cdot \frac{x}{2} \cdot \frac{x}{3}$

$R_A x - \frac{800x^2}{2} - \frac{wx^3}{6L} = M(x)$

$+2666.66x - 400x^2 - \frac{800x^3}{30} = M(x)$

$2666.66x - 400x^2 - 26.66x^3 = M(x)$

at $x = 0$, $M(x) = 0$

at $x = L = 5$, $M(5) = 0$

Max at 2.637.

Hence, $M(x=2.637) = 3761.5\ N\text{-}mm$

Shear at Section x-x

$\frac{wx}{L} = 160x.$

Shear at Section x-x

$+2666.66 + V(x) - \frac{wx}{L} \times \frac{x}{2} - 800 \times x$

$V(x) = -\frac{wx^2}{2L} + 2666.66 - (800 \times x)$

$V(x) = -\frac{800 \times x^2}{10} + 2666.66 - 4000$ (parabolic)

When $x = L = 5$, $V(x) = -3333.34$

To find where $V(x) = 0$,

$V(x) = \frac{wx^2}{2L} + 800 \times x - 2666.66$

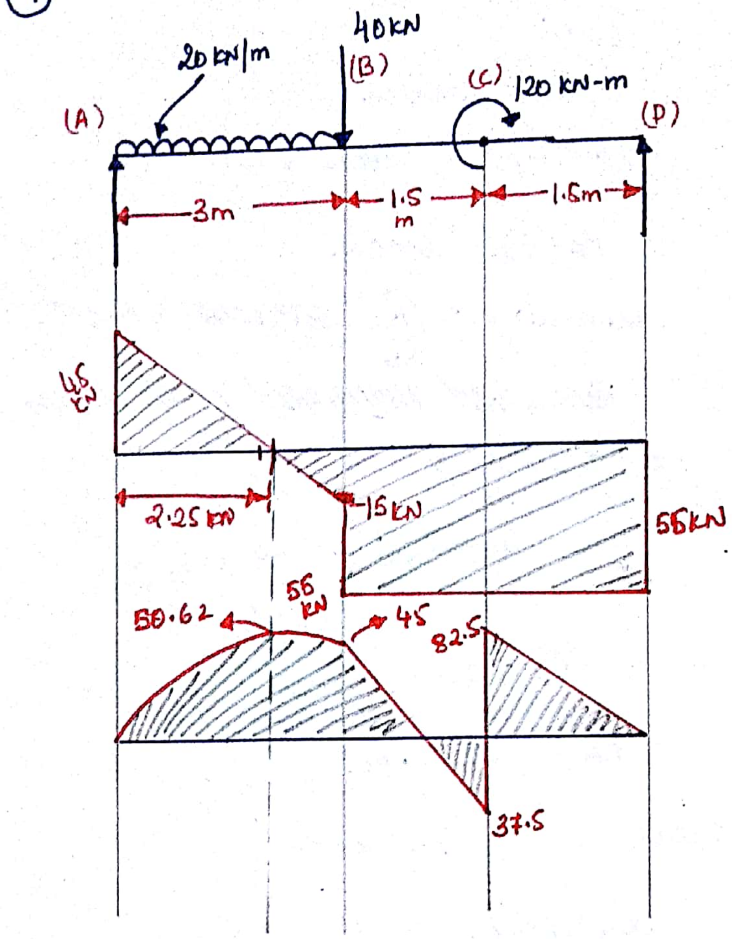
$\frac{wx^2}{2L} + 800x = 2666.66.$

$80x^2 + 800x = 2666.66$

$x^2 + 10x = 33.33$

$\boxed{x = 2.637}$

7



Reactions. $\Sigma M_A = 0$

$(20\times3\times1.5)+(40\times3)+(120) = R_D\times6$

$R_D = 55$ kN.

R_A; $\Sigma M_B = 0$

$-120 + (40\times3)+(20\times3\times4.5) = R_A\times6$

$R_A = 45$ kN.

Shear force at 'B'

$-20\times x + 45 = V(x)$

$x=0, \quad V(x) = 45$

$x=3 \quad V(x) = -15$

To find where $V(x)=0$,

at 2.25, $V(x) = 0$.

Calculate bending moment at 'B'

$45\times x - 20\times x\cdot\frac{x}{2} = 45x - 10x^2 = M(x)$

when $x=0, \quad M(x) = 0$

$x=3 \quad M(3) = 45$

$x=2.25 \quad M(2.25) = 50.625$

B.M @ C

$-20\times3(x-1.5) - 40\times(x-3) + 45\times x$, Linear Variation

$x=3 \quad M(x) = 45$ kN-m

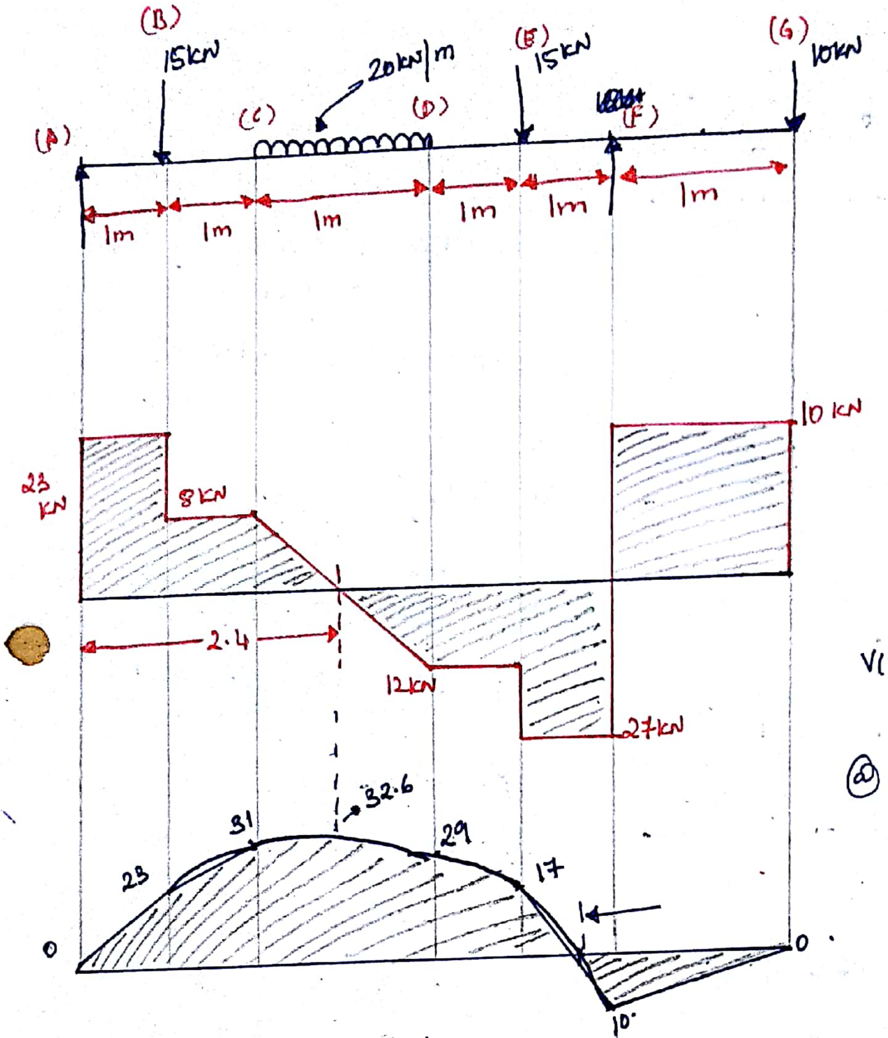
$x=4.5 \quad M(x) = -37.5$ kN. $\quad M(cn)=0$, when $x = 3.82$m.

B.M @ C-D.

$55\times x = M(x)$

$x=1.5 \cdot M(x) = 82.5$ kN-m

$x=0 \quad M(x) = 0.$



Reactions

$\Sigma M_A = 0$

$(15\times1) + (20\times1\times2.5) + (15\times4) + (10\times6) = R_F \times 5$

$R_F = 37$ kN

$R_A + R_F = 60$, $R_A = 23$ kN

Shear force at 'B' = 8 kN

$V(x) = 23 - 15 - 20(x-2)$

$= 8 - 20(x-2)$

@ $x = 2$ $V(x) = 8$

$x = 3$ $V(x) = -12$

$x = 2.4$ $V(x) = 0$

Bending Moment @ 'B' $M_{(B)} = 23\times1 = 23$ kN-m

$M(x) = (23\times x) + (-15\times(x-1)) + \left(-20(x-2)\frac{(x-2)}{2}\right)$

Solving, we get, $-10x^2 + 48x - 25 = M(x)$

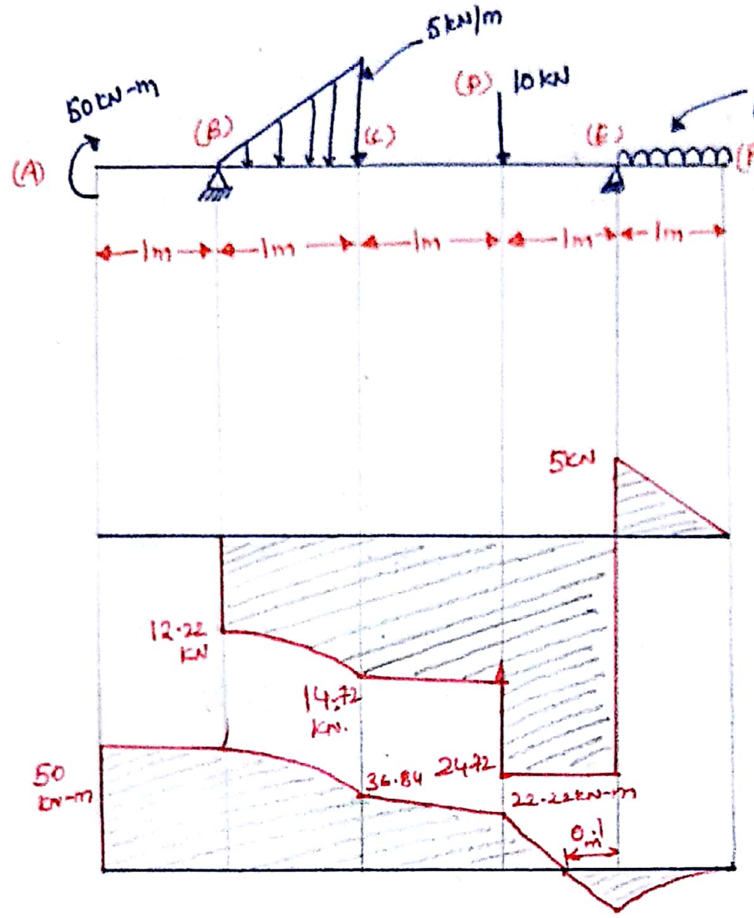
@ $x=2$, $M(x) = 31$ kN-m

@ $x=3$, $M(x) = 29$ kN-m

$M_{(E)} = (23\times4) - (15\times3) - (20\times1.5) = 17$ kN-m

$M_{(F)} = -10\times1 = -10$ kN-m

Scanned by CamScanner



Reactions :-

$R_B + R_E = 10 + (5 \times 1) + \frac{1}{2} \times 1 \times 5 =$

$10 + 5 + 2.5 = 17.5$ kN.

$\Sigma M_B = 0.$

$50 + (\frac{1}{2} \times 1 \times 5) \times (\frac{2}{3}) + 10 \times 2 + 5 \times 1 \times 3.5 = R_E \times 3$

$R_E = \dfrac{50 + 1.666 + 20 + 17.5}{3}$

$R_E = 29.72$ (+ve)

$R_B = 17.5 - 29.72$

$R_B = -12.22$ kN (-ve)

Shear force begins at 'B' because of pure bending @ 'A-B'.

$V_{(x)} = -12.22 - \frac{5x}{1} \cdot \frac{x}{2} = -12.22 - \frac{5}{2}x^2$

at $x = 0$ -12.22 kN. parabolic

@ $x = 1$ -14.72 kN (~~cubic~~ Variation).

@ E-F.

$5x = V(x)$

@ $x = 0$, $V(x) = 0$

$x = 1$ $V(x) = 5$.

Moment @ A = 50 kN-m

Moment @ C =

$50 - \frac{5x}{2} \cdot \frac{x \cdot x}{3}$ $50 - \frac{5x^3}{6} = M_{(c)}$

at $x = 0$ $M_{(x)} = 50$ kN-m

at $x = 1$ $M_{(x)} = 50 - 5/6 - 12.22 \times 1$

$M_{(c)} = 36.85$ kN-m.

$M_{(d)} = 50 - 12.22 \times 2 - \frac{5 \times 1}{2}(1 + 1/3) = 22.22$ kN-m

Similiarly @ $M_{(E)}$

$M_{(E)} = -5 \times \frac{1}{2} = -2.5$ kN

$M_{(x)} = 0 = 29.92 \times x - 5(1/2 + x)$

$-5x - 2.5 + 29.92x = 0$

$x = 0.1$ m.//.